



How data informs better decisions?

So far, you've discovered that there are many different ways data can be used. In our everyday lives, we use data when we wear a fitness tracker or read product reviews to make a purchase decision.

And in business, we use data to learn more about our customers, improve processes, and help employees do their jobs more effectively.

But this is just the tip of the iceberg.

One of the most powerful ways you can put data to work is with data-driven decision-making.

Data-driven decision-making is defined as using facts to guide business strategy.

Organizations in many different industries are empowered to make better, data-driven decisions by data analysts all the time.

The first step in data-driven decision-making is figuring out the business need.

Usually, this is a problem that needs to be solved.

For example, a problem could be a new company needing to establish better brand recognition, so it can compete with bigger, more well-known competitors.

Or maybe an organization wants to improve a product and needs to figure out how to source parts from a more sustainable or ethically responsible supplier.

Or, it could be a business trying to solve the problem of unhappy employees, low levels of engagement, satisfaction and retention.

Whatever the problem is, once it's defined, a data analyst finds data, analyzes it and uses it to uncover trends, patterns and relationships.

Sometimes the data-driven strategy will build on what's worked in the past.

Other times, it can guide a business to branch out in a whole new direction.

Let's look at a real-world example.

Think about a music or movie streaming service.

How do these companies know what people want to watch or listen to, and how do they provide it?

Well using data-driven decision-making, they gather information about what their customers are currently listening to, analyze it, then use the insights they've gained to make suggestions for things people will most likely enjoy in the future.

This keeps customers happy and coming back for more, which in turn means more revenue for the company.

Another example of data-driven decision-making can be seen in the rise of e-commerce.

It wasn't long ago that most purchases were made in a physical store, but the data showed people's preferences were changing.

So a lot of companies created entirely new business models that remove the physical store, and let people shop right from their computers or mobile phones with products delivered right to their doorstep.

In fact, data-driven decision-making can be so powerful, it can make entire business methods **obsolete**.

For example, data helped companies completely move away from corded phones and replace them with mobile phones.

By ensuring that data is built into every business strategy, data analysts play a critical role in their companies' success, but it's important to note that no matter how valuable data-driven decision-making is,

data alone will never be as powerful as data combined with human experience, observation, and sometimes even intuition.

To get the most out of data-driven decision-making, it's important to include insights from people who are familiar with the business problem.

These people are called subject matter experts, and they have the ability to look at the results of data analysis and identify any inconsistencies, make sense of gray areas, and eventually validate choices being made.

Organizations that work this way put data at the heart of every business strategy, but also benefit from the insights of their people.

It's a win-win.

As a data analyst, you play a key role in empowering these organizations to make data-driven decisions, which is why it's so important for you to understand how data plays a part in the decision-making process.

Explain this topic in simple terms

Data-driven decision-making is like being a detective solving a case, but instead of using clues, we use data!

Let's say a clothing store wants to know what new items to sell. Instead of just guessing, they can look at data like what's currently popular, what customers are searching for online, and even the weather to make informed decisions about what people are most likely to buy. This helps them stock the right items and keep their customers happy.

Give me a summary

This section of the "Foundations: Data, Data, Everywhere" course talks about **data-driven decision-making**.

Here are the key takeaways:

- **What it is:** Using data analysis to make informed business decisions.

- **How it works:** Identify a problem, gather data, analyze it to find trends and patterns, and use those insights to make decisions.
- **Benefits:** Can lead to better products, happier customers, and more successful businesses.
- **Examples:** Music streaming services using data to recommend songs, e-commerce platforms analyzing shopping habits, and companies using data to understand customer preferences.
- **Importance of human insight:** Data is most powerful when combined with human experience and intuition. Subject matter experts help validate data-driven decisions.

Give me real-life examples

You're right, real-life examples always help to understand a concept better! Let's dive into the world of ride-sharing with this example:

- **Imagine a company like Uber or Lyft wants to make sure enough drivers are available during peak hours.**
 - They can analyze data from past rides to see when and where demand is highest.
 - For example, they might find a **surge** in rides every weekday around 5 PM in the downtown area.
 - Using this data, they can offer incentives to drivers to encourage them to be on the road during those busy times, ensuring that riders can get rides when they need them.

This is just one example of how data-driven decision-making plays a role in our daily lives.

Give me practice questions

Okay, here are five questions to review the concepts from the lesson:

1. **True/False:** Data-driven decision-making relies solely on intuition and gut feelings.
2. **Multiple Choice:** Which of the following is NOT a step in the data-driven decision-making process?
 - Defining the problem
 - Collecting data

- Ignoring past trends
 - Analyzing data
3. **Open-Ended:** Can you think of a time when a data-driven decision might have helped you in your own life? Describe the situation and how data could have played a role.
 4. **Open-Ended:** Why is it important to combine data analysis with human insight when making decisions?
 5. **True/False:** Data-driven decision-making is only relevant for large corporations and not for small businesses or individuals.